

Six Sigma Project - Analyze Phase

Project Title

Reducing Average Handle Time in the Customer Support Team

Business Context

Our support team handles phone and chat inquiries for a small online retailer. Over the past six months, Average Handle Time (AHT) has increased from about 6 minutes to about 9 minutes, while customer satisfaction scores have dropped. This hurts customer experience and increases staffing costs.

Goal of the Analyze Phase

In the Measure phase we confirmed that AHT is high and unstable. In the Analyze phase, the goal is to understand why AHT is high and which causes matter most, using simple but structured tools: Cause & Effect analysis, graphical analysis, basic statistics, 8 wastes, Cost of Poor Quality, and FMEA. [Based on DMAIC Analyze guidance][web:601]

Summary of Key Findings by Deliverable

Cause & Effect / Why-Why

Using a simple fishbone (Ishikawa) diagram, the team grouped potential causes of high AHT into People, Process, Technology, and Policy. The Why-Why discussion highlighted three likely root causes:

- Inconsistent call handling steps between agents.
- Slow knowledge search in the help system.
- Too many transfers for billing and refund questions.

Histogram / Graphical Analysis

A histogram of AHT shows a long right tail: most calls are around 6-8 minutes, but a smaller group of calls takes 15-20 minutes. This suggests that a minority of “long” calls are pulling the average up. Histograms are commonly used in the Analyze phase to understand data distribution and variation. [web:598][web:605]

Process and Statistical Analysis

A simple process map shows the typical path: Welcome → Identify Request → Authenticate → Research / Resolve → Wrap-up. Time checks on 50 recent calls show that “Research / Resolve” and “Transfers” are where time is lost. Basic descriptive statistics confirm that calls with a transfer average about 4 minutes longer than calls without a transfer.

Eight Wastes (Value-Added Analysis)

Using the Lean “8 wastes” view (TIMWOODS), the team found clear examples of Waiting (agents waiting on slow systems), Overprocessing (double documentation), Motion (switching between many screens), and Skills waste (senior agents doing simple password resets). [web:599][web:549][web:546]

Cost of Poor Quality (COPQ)

Cost of Poor Quality was estimated by looking at:

- Extra agent time on long calls (internal failure cost).
- Lost sales due to abandoned calls (external failure cost).

If each extra minute costs ₹15 in staffing and about 2,000 calls per month run 3 minutes longer than needed, that alone is roughly ₹90,000 per month in avoidable internal cost. COPQ is the standard way to show how quality issues turn into real money. [web:600][web:602][web:604]

FMEA (Failure Modes and Effects Analysis)

A simple FMEA table was built for key steps in the call flow. For each step we listed failure modes (for example “Agent cannot find the correct answer”), effects (“Customer waits on hold”), and causes (“Knowledge base search is slow or unclear”), and then rated Severity, Occurrence, and Detection to get an RPN. High RPN items were:

- Knowledge base too slow or outdated.
- Transfers due to unclear ownership of billing issues.
- Inconsistent call wrap-up notes causing rework.

FMEA is commonly used in Six Sigma to prioritize risks and focus improvement work where it matters most. [web:594]

Summary of Next Steps

Based on the above tools, the Analyze phase points clearly to a few high-impact root cause themes: long “research” time, excessive transfers, and waste built into the process. The next steps focus on testing targeted improvements in the Improve phase (see separate “Summary of Next Steps” document).

Overarching Questions

a) Was there appropriate use of tools?

Yes. The project uses standard Analyze-phase tools: cause & effect analysis, graphical analysis, simple statistics, Lean waste analysis, COPQ, and FMEA. These are all widely recommended for finding root causes before jumping to solutions. [web:594][web:596]

b) Tools beyond the required ones

Beyond the required tools, a basic Pareto view of “reasons for long calls” was used informally to see which call types most often run over 15 minutes. This helped focus later analysis on billing and refund calls instead of all call types. Pareto analysis is a common, simple way to focus on the “vital few” issues. [web:594][web:598]

c) Is the project ready to move to Improve?

The project is ready to move to the Improve phase. We have:

- A clear problem: high AHT driven by a small group of very long calls.
- Plausible and verified causes: slow research, transfers, and process waste.
- Prioritized risks through FMEA and a first estimate of COPQ.

d) If not fully ready, what recovery measures are needed?

Before large-scale rollout, we still need:

- One more week of data that tags each call with “reason” and “transfer yes/no”.
- A small pilot test of a faster knowledge base search for 3–5 agents.

These actions can be done within 2–3 weeks and will reduce risk when changes are rolled out to the whole team.

e) Does the charter, problem, or scope need refinement?

Yes, the scope should be refined to focus first on billing and refund calls, because these show the highest AHT and most transfers. The new problem statement can be: “Reduce AHT for billing and refund calls by 2 minutes within three months, while maintaining customer satisfaction.” Narrowing the scope makes the Improve phase more realistic for a small team.